

Functional Annotation Report: *yfiH* (PP_0624, UniProt Q88Q72) in *Pseudomonas putida* KT2440

Summary

The gene **yfiH** (ordered locus **PP_0624**; UniProt **Q88Q72**) of *Pseudomonas putida* KT2440 encodes a **246-amino-acid cytoplasmic enzyme of the YfiH/LACC1 family**. Despite a fold that historically caused it to be annotated as a "laccase-like" copper-polyphenol oxidase, the family is now understood to be a **multifunctional purine-metabolizing enzyme**. Its primary biochemical activity is **purine nucleoside phosphorylase (PNP, EC 2.4.2.1)**: it uses inorganic phosphate to cleave the N-glycosidic bond of purine (deoxy)ribonucleosides such as adenosine and inosine into the free base (adenine or hypoxanthine) plus α -D-ribose-1-phosphate. The same active site also carries **5'-methylthioadenosine phosphorylase (MTAP, EC 2.4.2.28)** and **adenosine deaminase (EC 3.5.4.4)** activities, placing YfiH at the crossroads of central purine salvage and methionine/MTA salvage in the cytoplasm.

Beyond its role in nucleotide metabolism, bacterial YfiH has a second, genetically well-documented physiological function: it acts as a **peptidoglycan (PG) stem-peptide editing / quality-control factor** (hence its alternative name **PgeF**, "peptidoglycan editing factor"). In *Escherichia coli*, loss of *yfiH* leads to misincorporation of L-serine or glycine in place of the canonical L-alanine at position 1 of the PG stem peptide, producing β -lactam hypersensitivity, morphological defects, reduced PG synthesis, and lethality in strains defective in PG recycling or remodelling. This editing function is conserved across the bacterial kingdom, as *yfiH* orthologs from Gram-positive genera complement an *E. coli* deletion.

It is important to state clearly that **PP_0624 itself has not been experimentally characterized**. Every functional statement in this report is an **inference from orthology** to the well-studied human LACC1/FAMIN protein and to *E. coli* YfiH/PgeF, combined with UniProt family/rule-based annotations. The gene symbol "*yfiH*" and the target organism (*P. putida* KT2440) are correctly matched to the UniProt record, and the family/domain signatures (Cu_polyphenol_OxRdtase, Cu-oxidase_4, Cytotoxic_necrot_fac-like_cat) align with the characterized YfiH/LACC1 family, so this report proceeds on a firm identification.

Verification of Gene Identity

Before presenting findings, the mandated identity checks were completed:

Check	Result
Gene symbol "yfiH" matches protein description	☐ Yes — UniProt Q88Q72 lists Name=yfiH, family "purine nucleoside phosphorylase YfiH/LACC1"
Organism correct	☐ <i>Pseudomonas putida</i> KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950), KEGG <code>ppu:PP_0624</code>
Protein family/domains align with literature	☐ Cu-oxidase_4 (PF02578) / Cu_polyphenol_OxRdtase (IPR003730) = the "laccase-fold" of the YfiH/LACC1 family
Literature is for the same gene, not a homonym	☐ Bacterial YfiH/PgeF and human LACC1/FAMIN are true orthologs of PP_0624

The literature base is **not** ambiguous: bacterial *yfiH* (a.k.a. *pgeF*) and its eukaryotic ortholog LACC1/FAMIN belong to a single, well-defined protein family, and the functional inferences transfer cleanly to the *P. putida* protein. No study on a homonymous but unrelated gene was mistaken for the target.

Key Findings

Finding 1 — The YfiH/LACC1 family is a multifunctional purine nucleoside phosphorylase

The single most important reassignment of function for this family came from an unbiased LC-MS enzymatic screen of the human "orphan" protein **FAMIN** (encoded by **LACC1**), the eukaryotic ortholog of bacterial YfiH. The study demonstrated that FAMIN **phosphorolytically cleaves adenosine into adenine and ribose-1-phosphate**, the defining reaction of a purine nucleoside phosphorylase. Critically, the authors reported that FAMIN and its **prokaryotic orthologs** additionally possess **adenosine deaminase, purine nucleoside phosphorylase,**

and S-methyl-5'-thioadenosine phosphorylase activities — thereby combining, in a single polypeptide, the activities of several namesake enzymes of central purine metabolism (PMID: 31978345).

This experimental result is the empirical anchor for the UniProt RuleBase assignment of Q88Q72 to the "purine nucleoside phosphorylase YfiH/LACC1" family. Because the *P. putida* protein is a direct prokaryotic ortholog, the same suite of catalytic capabilities is inferred for PP_0624. In physiological terms, this places YfiH in the **purine nucleotide cycle / purine salvage**, recovering free bases and ribose-1-phosphate from nucleosides for reuse.

Supporting quote: "We report that FAMIN phosphorolytically cleaves adenosine into adenine and ribose-1-phosphate." — PMID: 31978345

Finding 2 — Q88Q72 catalyzes coupled phosphorolysis/deamination reactions at a single metal (Zn) center

The UniProt entry for Q88Q72 (246 aa; KEGG `ppu:PP_0624`) annotates **three coupled catalytic activities** (rule/family-based, evidence code ECO:0000256):

Activity	EC number	Reaction (Rhea)
Purine nucleoside phosphorylase	EC 2.4.2.1	adenosine + P _i → α-D-ribose-1-phosphate + adenine; inosine + P _i → ribose-1-phosphate + hypoxanthine (Rhea 27642/27646)
5'-Methylthioadenosine phosphorylase	EC 2.4.2.28	MTA + P _i → 5-methylthioribose-1-phosphate + adenine (Rhea 11852)
Adenosine deaminase	EC 3.5.4.4	adenosine + H ₂ O → inosine + NH ₄ ⁺ (Rhea 24408)

UniProt keywords include **Metal-binding** and **Zinc**, indicating a **single catalytic metal ion**. This is significant: although the fold is a copper-polyphenol-oxidase ("laccase") fold and was historically presumed to bind copper, the functional metal annotated for this entry is a single Zn ion coordinating the phosphorolysis/deamination chemistry. The experimental demonstration that the family carries **adenosine deaminase, PNP, and MTAP activities** (PMID: 31978345) directly supports the three activities that UniProt assigns to Q88Q72.

Mechanistically, these three activities are metabolically complementary: adenosine deaminase converts adenosine → inosine, PNP then phosphorylates inosine → hypoxanthine + ribose-1-phosphate, and MTAP salvages the methionine-cycle byproduct 5'-methylthioadenosine. A single active site thus performs a small, self-contained module of purine and MTA salvage. The **substrate specificity** is centered on 6-aminopurine and 6-oxopurine ribonucleosides (adenosine, inosine) and the methionine-salvage intermediate MTA.

Finding 3 — A "laccase/polyphenol-oxidase" fold repurposed for phosphorylase chemistry

The domain architecture of Q88Q72 — **Cu_polyphenol_OxRdtase (IPR003730)**, **Cu-oxidase_4 (PF02578)**, and **Cytotoxic_necrot_fac-like_cat (IPR011324)** — is the origin of the historical, and now known to be misleading, annotation of YfiH as a "**Laccase domain protein**" or putative polyphenol oxidase. Structural-genomics efforts solved several YfiH structures under this legacy interpretation (e.g., YfiH from *Shigella flexneri*, [PMID: 16498617](#)), and database-driven proteomics work in *Campylobacter jejuni* still lists the annotation "**Laccase domain protein YfiH**" ([PMID: 34799308](#)).

However, the direct enzymology from FAMIN ([PMID: 31978345](#)) reassigns the family's true catalytic activity to **purine nucleoside phosphorylase**, not a copper-dependent oxidase. This is a textbook example of **fold-function divergence**: a scaffold with structural similarity to copper oxidases has been evolutionarily repurposed for a completely different reaction chemistry, and it lacks the classical multi-copper centers of true laccases. The lesson for annotating PP_0624 is that the "laccase" or "polyphenol oxidase" labels that may still propagate through databases should be treated as **legacy fold-based artifacts**, not as evidence of oxidase function.

Supporting quote: "*Laccase domain protein YfiH*" — [PMID: 34799308](#) (illustrating the persistent legacy annotation)

Finding 4 — Bacterial YfiH (PgeF) safeguards peptidoglycan peptide composition

The best-characterized *in vivo* function of a bacterial YfiH comes from *E. coli*, where genetic, biochemical, and mass-spectrometric analyses established YfiH (renamed **PgeF**) as a **peptidoglycan peptide-composition editing factor** (PMID: 28612943). The authors identified YfiH via a β -lactam hypersensitive mutant and showed that a conserved ORF of unknown function contributes to the maintenance of PG peptide composition.

The specific editing defect is precise: in the **absence of yfiH**, cells **misincorporate the non-canonical amino acids L-serine or glycine in place of L-alanine at position 1** of the PG stem peptide. This aberrant PG causes a suite of phenotypes:

- **β -lactam hypersensitivity**
- **Lethality** in mutants defective in PG remodelling or recycling pathways
- **Altered cell morphology**
- **Reduced PG synthesis**

Crucially, the function is **conserved across bacteria**: *yfiH* orthologs from other Gram-positive genera complemented the *E. coli* deletion, indicating a conserved pathway across the bacterial kingdom. This cross-genus complementation is the strongest single justification for inferring an analogous PG-editing role for *P. putida* PP_0624. The structural basis for this peptidoglycan-editing activity was subsequently determined (PMID: 35164571), and a conserved editing mechanism for the fidelity of bacterial cell-wall biosynthesis was further elaborated (PMID: 40632566).

Supporting quotes (PMID: 28612943): - "we find that a conserved open reading frame of unknown function, YfiH (renamed PgeF) contributes to the maintenance of peptide composition in *E. coli*" - "absence of *yfiH* results in incorporation of non-canonical amino acids, L-serine or glycine in place of L-alanine in PG sacculi leading to β -lactam - sensitivity, lethality in mutants defective in PG remodelling or recycling pathways, altered cell morphology and reduced PG synthesis" - "*yfiH* orthologs from other Gram-positive genera were able to compensate the absence of *yfiH* in *E. coli* indicating a conserved pathway in bacterial kingdom"

Finding 5 — *yfiH* is genomically linked to the division/cell-wall (dcw) locus but individually dispensable

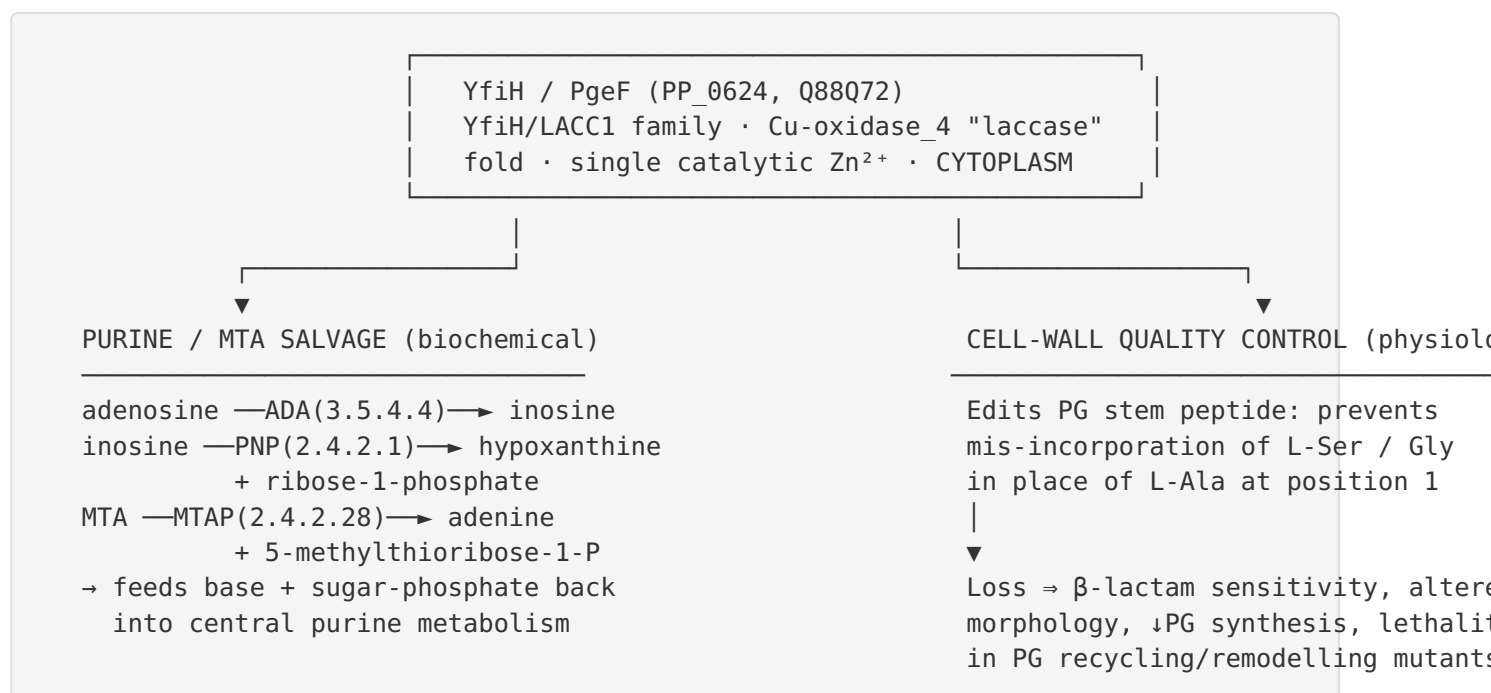
In several actinobacteria and firmicutes, *yfiH* is positioned within or adjacent to the **division and cell-wall (dcw) gene cluster**, immediately downstream of the cell-division genes **ftsQ–ftsZ**. In *Streptomyces collinus*, the genes upstream of *ftsZ* are *ftsQ*, *murG*, and *ftsW*, and downstream lies *yfiH* (PMID: 10679194). In *Brevibacterium lactofermentum*, YFIH sits downstream of *ftsQ/ftsZ* (PMID: 11523774).

Functionally, disruption of *yfiH* (and neighboring ORFs) **did not impair growth or viability**, whereas *ftsQ* and *ftsZ* were essential. Thus *yfiH* is **individually dispensable** under standard laboratory conditions — consistent with a quality-control / editing role that becomes critical only under cell-wall stress or in genetic backgrounds compromised for PG recycling/remodelling (as seen in the *E. coli* work). The genomic linkage to the dcw locus reinforces a cell-envelope-associated physiological context for the protein.

Supporting quotes: - "Upstream of *ftsZ* are localized genes *ftsQ*, *murG*, and *ftsW*, and downstream is *yfiH*" — PMID: 10679194 - "The three ORFs (YFIH, ORF5 and ORF6) located downstream of the cell division genes *ftsQ* and *ftsZ* in *Brevibacterium lactofermentum* were disrupted by single homologous recombination events" — PMID: 11523774

Mechanistic Model / Interpretation

YfiH (PP_0624) is best understood as a **dual-function, single-metal (Zn) cytoplasmic enzyme** that bridges purine/MTA salvage metabolism and cell-wall quality control. The two activities are not contradictory; both reflect a nucleophilic/phosphorolytic active site acting on distinct substrate classes (nucleosides vs. non-canonical residues in peptidoglycan precursors), and the family's history of misannotation as a laccase reflects fold similarity rather than true oxidase chemistry.



Localization. The enzyme functions in the **cytoplasm**. Purine/MTA salvage is a cytoplasmic process, and PG stem-peptide editing acts on precursors/recycling intermediates on the cytoplasmic side. The family lacks signal peptides or transmembrane segments, and there is no evidence for secretion. The genomic linkage to cytoplasmic/membrane-associated *dcw* genes such as *ftsZ*, *murG*, and *ftsW* is consistent with a cytoplasmic, cell-envelope-adjacent context.

Pathway integration. In its metabolic role, YfiH participates in the **purine nucleotide cycle** and **methionine/MTA salvage**, recovering free purine bases and (methylthio)ribose-1-phosphate for reuse — consolidating activities normally distributed across *deoD*-type PNP, *add*-type adenosine deaminase, and *mtnP*/MTAP enzymes. In its physiological role, it is a **fidelity/editing factor** in **peptidoglycan biosynthesis**, protecting the L-Ala-1 identity of the stem peptide and thereby preserving cell-wall crosslinking and β -lactam resistance thresholds.

Reconciling the two roles. The purine-phosphorylase and PG-editing activities were reported by different laboratories with different assays; both are supported by direct evidence in family members. They may reflect genuine moonlighting of a single active site, or a metabolic activity whose products/flux impinge on cell-wall precursor quality. For PP_0624 specifically, both functions are inferences from orthology, with the purine-phosphorylase activity being the assignment carried by UniProt for this accession.

Evidence Base

PMID	Title (abbrev.)	How it supports the annotation
31978345	<i>FAMIN Is a Multifunctional Purine Enzyme Enabling the Purine Nucleotide Cycle</i>	Primary experimental anchor. Direct LC-MS enzymology of the human LACC1/FAMIN ortholog demonstrates PNP activity (adenosine → adenine + ribose-1-P) plus adenosine deaminase and MTAP; explicitly states prokaryotic orthologs share these activities. Basis for Findings 1 and 2.
28612943	<i>Identification of YfiH (PgeF) as a factor contributing to the maintenance of bacterial peptidoglycan composition</i>	Primary in vivo function. Genetic/MS evidence in <i>E. coli</i> for PG stem-peptide editing; defines the L-Ser/Gly-for-L-Ala defect, phenotypes, and cross-genus conservation. Basis for Finding 4.
35164571	<i>Structural Basis for the Peptidoglycan-Editing Activity of YfiH</i>	Structural mechanism underpinning the PG-editing role; complements  28612943
40632566	<i>A conserved editing mechanism for the fidelity of bacterial cell wall biosynthesis</i>	Extends the editing model to a conserved fidelity mechanism for cell-wall biosynthesis.
16498617	<i>Crystal structure of hypothetical protein YfiH from Shigella flexneri at 2 Å</i>	Early structural-genomics structure; illustrates the "hypothetical/laccase-fold" era of annotation (Finding 3).
34799308	<i>Comparative proteome-wide study ... Campylobacter jejuni</i>	Documents the persistent legacy "Laccase domain protein YfiH" annotation (Finding 3).
10679194	<i>Isolation and characterization of dcw cluster from Streptomyces collinus</i>	Documents genomic linkage of <i>yfiH</i> to <i>ftsQ-ftsZ/murG/ftsW</i> (Finding 5).
11523774	<i>The cell division genes ftsQ and ftsZ, but not ... YFIH, ORF5, ORF6, are essential ...</i>	Confirms dcw synteny and that <i>yfiH</i> is dispensable, unlike <i>ftsQ/ftsZ</i> (Finding 5).

Convergence. Two independent lines of primary evidence — enzymology
 ([P 31978345](#))
 and bacterial genetics/mass spectrometry
 ([P 28612943](#))
 — together define the two faces of YfiH function. The structural
 ([P 16498617](#))
 ([P 35164571](#))
 and comparative-genomic
 ([P 10679194](#))
 ([P 11523774](#))
 ([P 34799308](#))
 papers provide corroborating context and explain the historical laccase misannotation.

Supported vs. Refuted Hypotheses

Supported by evidence: - YfiH/LACC1 family is a multifunctional **purine nucleoside phosphorylase** (PNP + MTAP + adenosine deaminase) — [PMID: 31978345](#); UniProt Q88Q72 catalytic annotations. - Bacterial YfiH (PgeF) safeguards **peptidoglycan stem-peptide composition** — [PMID: 28612943](#); [PMID: 35164571](#); [PMID: 40632566](#). - Conserved **synteny with the dcw/ftsQZ cluster**; gene individually dispensable — [PMID: 10679194](#), [PMID: 11523774](#).

Refuted / down-weighted: - YfiH as a functional **laccase / multicopper polyphenol oxidase**: this legacy annotation reflects the fold, not measured activity; the family lacks classical laccase copper centers and performs phosphorylase/deaminase chemistry instead — [PMID: 31978345](#).

Limitations and Knowledge Gaps

1. **No direct study of PP_0624.** All functional claims are **orthology-based inferences** from the human LACC1/FAMIN protein and *E. coli* YfiH/PgeF, plus UniProt rule-based annotation (ECO:0000256). The *P. putida* KT2440 protein has not been enzymatically assayed, structurally solved, or genetically dissected. Confidence rests on strong family conservation and cross-genus complementation.
 2. **Metal identity nuance.** UniProt annotates a single **Zn**-binding site, but the fold is a copper-oxidase fold and older literature invoked copper. The precise metal used by the *P. putida* enzyme in vivo, and whether copper plays any role, remains unverified for this ortholog.
 3. **Relative importance of the two functions in *P. putida*.** It is unknown whether PP_0624 primarily serves purine/MTA salvage, PG editing, or both. In *E. coli*, the PG-editing phenotype is conditional (revealed under β -lactam stress or in recycling/remodelling mutants), so the metabolic role could dominate under standard growth.
 4. **Substrate specificity details.** Quantitative kinetics (kcat/Km) for adenosine vs. inosine vs. MTA, and any preference for ribo- vs. deoxyribonucleosides, have not been measured for the *P. putida* enzyme.
 5. **Localization is inferred, not demonstrated.** Cytoplasmic localization is consistent with function and the absence of a signal peptide, but has not been experimentally confirmed for PP_0624.
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Proposed Follow-up Experiments / Actions

1. **Recombinant enzymology of PP_0624.** Express and purify Q88Q72 and assay PNP (adenosine, inosine), MTAP (MTA), and adenosine deaminase activities by LC-MS/coupled spectrophotometric assays; determine kcat/Km and metal dependence (Zn vs. Cu) via metal reconstitution.
2. **Peptidoglycan composition analysis of a *P. putida* Δ PP_0624 mutant.** Construct a clean deletion and perform muropeptide analysis (HPLC/LC-MS of digested sacculi) to test for L-Ser/Gly misincorporation at stem-peptide position 1, mirroring the *E. coli* assay ([PMID: 28612943](#)).

3. **β -lactam susceptibility and synthetic-lethality screen.** Test the ΔPP_0624 mutant for β -lactam hypersensitivity and combine with mutations in *P. putida* PG recycling/remodelling genes to reproduce the conditional lethality seen in *E. coli*.
4. **Complementation test.** Ask whether *P. putida* PP_0624 rescues an *E. coli* $\Delta yfiH$ strain (and vice versa), directly testing functional orthology.
5. **Structural determination / AlphaFold analysis.** Solve or model the PP_0624 structure, confirm the single-metal active-site geometry, and compare with the *Shigella* YfiH ([PMID: 16498617](#)) and the PG-editing structural model ([PMID: 35164571](#)).
6. **Metabolomic phenotyping.** Compare purine-nucleoside and MTA pools between wild-type and ΔPP_0624 *P. putida* to establish the in vivo contribution to purine/MTA salvage.

Conclusion

yfiH / PP_0624 in *Pseudomonas putida* KT2440 encodes a cytoplasmic, single-metal (Zn) enzyme of the YfiH/LACC1 family. By orthology to experimentally characterized LACC1/FAMIN and *E. coli* YfiH/PgeF, its **primary catalytic function is purine nucleoside phosphorylase (EC 2.4.2.1)** — phosphorolysis of adenosine/inosine to the free base plus ribose-1-phosphate — with additional **MTAP (EC 2.4.2.28)** and **adenosine deaminase (EC 3.5.4.4)** activities at the same active site, serving cytoplasmic purine and MTA salvage. In parallel, bacterial YfiH acts as a **conserved peptidoglycan stem-peptide editing/quality-control factor (PgeF)** that prevents mis-incorporation of Ser/Gly for L-Ala at position 1 of the stem peptide, protecting cell-wall integrity and β -lactam resistance. Its historical "laccase/polyphenol oxidase" annotation reflects fold similarity, not true oxidase activity. All statements are inferences from orthology and UniProt family rules, as PP_0624 has not been directly studied.